BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

Blood flow of the abdomen and lower limb

The written notes for the blood flow lecture: the abdominal aorta and its branches, the inferior vena cava and its tributaries, the hepatic portal system, and the vessels and nerves of the lower limb. Module 4 competencies 126 and 127, with nine worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Competency 126. The abdominal aorta and its branches

In your course notes: Module 4 packet, Chapter 5, competency 126 (page 46).

Dr. Rennie's video for this part: No concept video for this chapter. Use the flow charts in the Module 4 lab packet and the cadaver.

You should be able to: Identify the major branches of the abdominal aorta and the vessels of the hepatic portal system on the cadaver.

The abdominal aorta is the aorta below the diaphragm. It enters the abdomen through the aortic hiatus at about T12, runs down the front of the lumbar vertebrae just left of the midline with the inferior vena cava on its right, and ends at about L4 by dividing into the right and left common iliac arteries. Its branches come in two groups: unpaired branches leave the front and feed the digestive organs; paired branches leave the sides and feed the kidneys, the gonads, and other structures that come in twos.

Branch	About where it leaves	What it supplies
Celiac trunk (unpaired)	T12, just below the diaphragm	Short; splits almost at once into the left gastric, splenic, and common hepatic arteries
Left gastric artery	From the celiac trunk	The stomach along the lesser curvature and the lower esophagus
Splenic artery	From the celiac trunk	Runs left along the top of the pancreas to the spleen, with branches to the pancreas and stomach
Common hepatic artery	From the celiac trunk	Runs right; gives the hepatic artery proper to the liver and gallbladder, plus branches to the stomach, duodenum, and pancreas
Superior mesenteric artery (unpaired)	L1	Runs in the mesentery to most of the small intestine, part of the pancreas, the cecum, appendix, ascending colon, and most of the transverse colon
Renal arteries (paired)	L1 to L2	The kidneys
Gonadal arteries (paired)	L2	The testes (testicular arteries) or the ovaries (ovarian arteries)
Inferior mesenteric artery (unpaired)	L3	The end of the transverse colon, the descending colon, the sigmoid colon, and the upper rectum
Right and left common iliac arteries	L4, where the aorta ends	The pelvis and lower limbs, through the internal and external iliac arteries

- The three gut branches go in the same order as the gut. Celiac feeds the top (stomach, liver, spleen, pancreas, start of the duodenum). Superior mesenteric feeds the middle (small intestine and the colon through most of the transverse). Inferior mesenteric feeds the end (the rest of the colon and the upper rectum).
- The three unpaired branches are the celiac trunk, superior mesenteric, and inferior mesenteric. Renal and gonadal arteries are paired.
- The hepatic artery proper rides up in the free edge of the lesser omentum to the porta hepatis.

Arteries of the pelvis

- Each common iliac artery divides in front of the sacroiliac joint. The internal iliac artery dives into the pelvis and supplies the bladder, rectum, reproductive organs, pelvic walls, gluteal muscles, and external genitalia. The external iliac artery runs along the psoas major, passes under the inguinal ligament, and becomes the femoral artery.
- The medial umbilical ligament is a fibrous cord on the inside of the anterior abdominal wall running from the pelvis toward the umbilicus. It is what remains of the fetal umbilical artery, a branch of the internal iliac. The part nearest the internal iliac stays open and supplies the bladder.

Competency 126. The inferior vena cava and the veins of the abdomen

In your course notes: Module 4 packet, Chapter 5, competency 126 (page 48).

The inferior vena cava (IVC) forms at about L5 where the right and left common iliac veins join. It runs up the right side of the aorta, passes through the central tendon of the diaphragm at about T8, and empties into the right atrium.

Vein	Drains	Empties into
External iliac vein	The lower limb; it is the femoral vein above the inguinal ligament	Common iliac vein
Internal iliac vein	The pelvis	Common iliac vein
Right and left common iliac veins	The pelvis and lower limb	Join to form the inferior vena cava
Renal veins	The kidneys. The left renal vein is longer because it crosses in front of the aorta to reach the IVC	Inferior vena cava
Right gonadal vein	The right testis or ovary	Inferior vena cava
Left gonadal vein	The left testis or ovary	Left renal vein
Hepatic veins	The liver, after the sinusoids and central veins	Inferior vena cava, just below the diaphragm
Inferior vena cava	Most of the body below the diaphragm	Right atrium

- The gonadal veins are not symmetric. Right into the IVC, left into the left renal vein.
- The stomach, intestines, spleen, and pancreas are not on this list. Their blood goes to the liver first, through the hepatic portal system.

Competency 126. The hepatic portal system

In your course notes: Module 4 packet, Chapter 5, competency 126 (page 48), and Chapter 4, competency 122 (page 41).

A portal system is a route in which blood passes through two capillary beds in a row before it returns to the heart. In the hepatic portal system the first bed is in the walls of the stomach, intestines, spleen, and pancreas, and the second bed is the sinusoids of the liver. The hepatic portal vein connects them. Everything absorbed from the gut, nutrients and whatever came with them, is processed by the liver before it reaches the rest of the body. It is also why cancers of the digestive tract so often spread to the liver first.

Vessel	Drains	Empties into
Superior mesenteric vein	The small intestine, cecum, ascending colon, and most of the transverse colon; runs beside the superior mesenteric artery	Joins the splenic vein to form the hepatic portal vein
Inferior mesenteric vein	The descending colon, sigmoid colon, and upper rectum	Usually the splenic vein
Splenic vein	The spleen, pancreas, and part of the stomach; runs along the back of the pancreas	Joins the superior mesenteric vein to form the hepatic portal vein
Hepatic portal vein	All of the above. Forms behind the neck of the pancreas, runs up in the free edge of the lesser omentum, and enters the liver at the porta hepatis	The hepatic sinusoids
Hepatic veins	The liver, from the central veins	Inferior vena cava

A glucose molecule, from a villus to the heart

1. Capillaries of an intestinal villus in the jejunum. The glucose enters the blood here: the first capillary bed.
2. Small veins of the intestinal wall and mesentery.
3. Superior mesenteric vein.
4. Hepatic portal vein, formed where the superior mesenteric and splenic veins join.
5. Porta hepatis, then a portal vein branch in a portal triad.
6. Hepatic sinusoids, past the hepatocytes: the second capillary bed.

7. Central vein of the lobule.
8. Hepatic veins, out of the back of the liver.
9. Inferior vena cava, through the diaphragm at T8.
10. Right atrium.

- The hepatic portal vein carries blood into the liver. The hepatic veins carry it out. Portal in, hepatic out.
- Two things enter the porta hepatis (the portal vein and the hepatic artery proper) and the hepatic ducts leave it. The hepatic veins do not use the porta hepatis; they leave the posterior surface straight into the vena cava.
- Portal vein blood is nutrient-rich and oxygen-poor. The hepatic artery brings the oxygen. They mix in the sinusoids and flow in toward the central vein, while bile flows out toward the triads.

Competency 127. Arteries and veins of the lower limb

In your course notes: Module 4 packet, Chapter 5, competency 127 (page 49).

You should be able to: Identify the femoral and lower-limb blood vessels and the major nerves of the abdomen and lower limb on the cadaver.

In the lower limb, arteries, veins, and nerves run together in bundles between the muscles you already know. Use the muscles as landmarks. When you name a vessel, say right or left and artery or vein.

One artery, renamed at each landmark

1. External iliac artery: along the pelvic brim from the common iliac.
2. Femoral artery: the same vessel after it passes under the inguinal ligament. Its pulse is felt about halfway between the anterior superior iliac spine and the pubic symphysis. It runs through the femoral triangle, gives off the deep femoral artery to most of the thigh muscles, then runs down under the sartorius.
3. Popliteal artery: the same vessel after it passes through the adductor hiatus in adductor magnus to reach the back of the knee.
4. Anterior and posterior tibial arteries: the popliteal divides just below the knee. The anterior tibial goes forward into the front of the leg and becomes the dorsalis pedis artery on the top of the foot. The posterior tibial, the larger branch, runs down the back of the leg deep to the gastrocnemius and soleus beside the tibial nerve, gives off the fibular artery to the lateral leg, passes behind the medial malleolus (second pulse point), and divides into the plantar arteries in the sole.

Femoral triangle

The space just below the inguinal ligament, bounded by the inguinal ligament above, the sartorius laterally, and the adductor longus medially. From lateral to medial: femoral nerve, femoral artery, femoral vein, empty space, lymphatics (NAVEL). The vein is medial to the artery.

Adductor hiatus

The opening in adductor magnus through which the femoral artery and vein pass to the back of the knee, where their names change to popliteal.

Veins of the lower limb

Vessel	Deep or superficial	Course and landmarks
Posterior and anterior tibial veins, popliteal vein, femoral vein	Deep	Run beside the arteries and share their names. The tibial veins join into the popliteal vein, which becomes the femoral vein up the thigh (medial to the artery in the triangle), which becomes the external iliac vein above the inguinal ligament. The deep veins carry most of the blood out of the limb
Great saphenous vein	Superficial	The longest vein in the body. Starts on the medial side of the top of the foot, passes in front of the medial malleolus, runs up the medial leg and thigh in the fat under the skin, and empties into the femoral vein just below the inguinal ligament. Surgeons use it as a graft in coronary bypass surgery
Small saphenous vein	Superficial	Up the back of the calf into the popliteal vein

- The name changes at three landmarks on the way to the foot: the diaphragm (thoracic to abdominal aorta), the inguinal ligament (external iliac to femoral), and the adductor hiatus (femoral to popliteal).
- The great saphenous vein is superficial and medial. The femoral vein is the deep one.

Competency 127. Nerves of the abdomen and lower limb

In your course notes: Module 4 packet, Chapter 5, competency 127 (page 50).

Only the femoral nerve is on your Module 4 structure list, and the plexus notes also name the obturator nerve. The rest are here so you can recognize what you see on the posterior abdominal wall and in the limb. You identify nerves by location; you are not asked which nerve supplies each muscle.

On the posterior abdominal wall

The lumbar plexus forms inside the psoas major from the anterior rami of L1 to L4. Its branches come out of the psoas and cross the posterior wall, which is where you see them on the cadaver.

Nerve	Spinal nerves	Where you see it	What it supplies
Iliohypogastric	L1	Crossing the posterior wall toward the iliac crest	Lower abdominal wall muscles, skin above the pubis
Ilioinguinal	L1	Just below the iliohypogastric, then through the inguinal canal	Lower abdominal wall, skin of the upper medial thigh and external genitalia
Genitofemoral	L1 to L2	Down the front of the psoas major	Skin of the external genitalia and a patch of upper anterior thigh
Lateral femoral cutaneous	L2 to L3	Across the iliacus toward the anterior superior iliac spine	Skin of the lateral thigh
Femoral	L2 to L4	Out of the lateral side of the psoas, in the groove between psoas major and iliacus	Anterior thigh muscles and skin
Obturator	L2 to L4	Out of the medial side of the psoas, to the obturator foramen	Adductor muscles and skin of the medial thigh

- Most of the anterolateral abdominal wall is supplied by the lower intercostal nerves (T7 to T11) and the subcostal nerve (T12); the lumbar plexus supplies only the lowest strip above the pubis and the groin.
- The anterior and posterior vagal trunks (vagus, cranial nerve X) enter the abdomen on the esophagus through the esophageal hiatus and supply the stomach and other abdominal organs.

In the lower limb

Nerve	Plexus	Course	Muscles supplied
Femoral nerve	Lumbar plexus (L2 to L4)	Under the inguinal ligament, lateral to the femoral artery, into the femoral triangle	Anterior thigh: quadriceps femoris and sartorius; a branch continues to the skin of the medial leg
Sciatic nerve	Sacral plexus (L4 to S3)	Out the greater sciatic foramen, deep to gluteus maximus, down the back of the thigh deep to the hamstrings; the largest nerve in the body	The hamstrings; splits above the knee into the tibial and common fibular nerves
Tibial nerve	Sacral, from the sciatic	Straight down through the back of the knee and the posterior leg deep to gastrocnemius and soleus, beside the posterior tibial artery, behind the medial malleolus into the sole	Posterior leg muscles and the sole of the foot
Common fibular nerve	Sacral, from the sciatic	Along the biceps femoris tendon to the lateral knee, around the neck of the fibula just under the skin, where it splits into superficial and deep branches	Superficial branch: lateral leg (fibularis longus and brevis). Deep branch: anterior leg (tibialis anterior, extensor hallucis longus, extensor digitorum longus, fibularis tertius)

- The tibial and common fibular nerves come off the sciatic nerve, not straight off the plexus.
- Because the common fibular nerve lies against the neck of the fibula just under the skin, a blow or a tight cast there can injure it and weaken the muscles that lift the foot: foot drop.

Worked questions

Nine questions in the style of the lecture exam. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 126

Which of these is an unpaired branch of the abdominal aorta?

A. Renal artery

B. Gonadal artery

C. Superior mesenteric artery

D. Common iliac artery

Answer: C. The superior mesenteric artery leaves the front of the aorta at about L1 as a single vessel and feeds the middle of the gut. The three unpaired branches are the celiac trunk, the superior mesenteric, and the inferior mesenteric.

Why the others are wrong

- **A** The renal arteries are paired, one to each kidney, leaving the sides of the aorta.
- **B** The gonadal arteries are paired, one to each testis or ovary.
- **D** The common iliac arteries are the paired terminal branches where the aorta ends at L4.

QUESTION 2 · COMPETENCY 126

The celiac trunk splits into which three arteries?

A. Left gastric, splenic, common hepatic

B. Superior mesenteric, renal, gonadal

C. Left gastric, right gastric, inferior mesenteric

D. Splenic, renal, common hepatic

Answer: A. The celiac trunk is short and splits almost at once into the left gastric artery (stomach and lower esophagus), the splenic artery (spleen, pancreas, stomach), and the common hepatic artery (liver, gallbladder, stomach, duodenum, pancreas).

Why the others are wrong

- **B** Those are separate branches of the aorta itself, below the celiac trunk.
- **C** The inferior mesenteric artery is its own unpaired branch at L3, not part of the celiac trunk.
- **D** The renal arteries come straight off the sides of the aorta, not from the celiac trunk.

QUESTION 3 · COMPETENCY 126

The left gonadal vein empties into the

A. inferior vena cava

B. left renal vein

C. left common iliac vein

D. hepatic portal vein

Answer: B. The gonadal veins are not symmetric. The right one drains straight into the inferior vena cava; the left one drains into the left renal vein, which then crosses in front of the aorta to the IVC.

Why the others are wrong

- **A** That is the right gonadal vein.
- **C** The common iliac veins collect from the pelvis and lower limb, below the level of the gonadal veins.
- **D** The hepatic portal vein drains the digestive organs and spleen, not the gonads.

QUESTION 4 · COMPETENCY 126

What makes the hepatic portal system a portal system?

A. It carries oxygen-rich blood to the liver

B. Its blood passes through two capillary beds in a row before returning to the heart

C. It drains directly into the right atrium

D. It is the only vein that carries blood away from the liver

Answer: B. A portal system is two capillary beds in series. The first is in the wall of the gut, spleen, and pancreas; the second is the hepatic sinusoids. The hepatic portal vein connects them.

Why the others are wrong

- **A** Portal vein blood is oxygen-poor and nutrient-rich. The hepatic artery brings the oxygen.
- **C** The portal vein empties into the liver sinusoids. The hepatic veins and the inferior vena cava carry the blood on to the right atrium.
- **D** The portal vein carries blood into the liver. The hepatic veins carry it away.

QUESTION 5 · COMPETENCY 126

Which vessel carries blood out of the liver?

A. Hepatic portal vein

B. Hepatic artery proper

C. Hepatic veins

D. Splenic vein

Answer: C. The central veins of the lobules join into the hepatic veins, which leave the back of the liver and empty into the inferior vena cava just under the diaphragm.

Why the others are wrong

- **A** The hepatic portal vein brings nutrient-rich blood from the gut into the liver at the porta hepatis.
- **B** The hepatic artery proper brings oxygen-rich blood into the liver.
- **D** The splenic vein drains the spleen, pancreas, and part of the stomach and joins the superior mesenteric vein to form the portal vein, upstream of the liver.

QUESTION 6 · COMPETENCY 127

The femoral artery becomes the popliteal artery when it

A. passes under the inguinal ligament

B. gives off the deep femoral artery

C. passes through the adductor hiatus to the back of the knee

D. divides below the knee

Answer: C. The femoral artery runs down under the sartorius and through the adductor hiatus, the opening in adductor magnus, to reach the popliteal fossa. From there it is the popliteal artery.

Why the others are wrong

- **A** At the inguinal ligament the external iliac artery becomes the femoral artery, one name change earlier.
- **B** The deep femoral branch leaves in the femoral triangle, but the femoral artery keeps its name and continues down the thigh.
- **D** Below the knee the popliteal artery divides into the anterior and posterior tibial arteries.

QUESTION 7 · COMPETENCY 127

In the femoral triangle, from lateral to medial, the order is

A. vein, artery, nerve

B. artery, vein, nerve

C. nerve, artery, vein

D. nerve, vein, artery

Answer: C. NAVEL: nerve, artery, vein, empty space, lymphatics, from lateral to medial. The femoral vein lies on the medial side of the artery.

Why the others are wrong

- **A** This is the reverse order; the nerve is the most lateral structure.
- **B** The nerve is lateral to the artery, not medial to the vein.
- **D** The artery lies between the nerve and the vein.

QUESTION 8 · COMPETENCY 127

Which vein is superficial and runs up the medial side of the leg and thigh, passing in front of the medial malleolus?

A. Femoral vein

B. Popliteal vein

C. Great saphenous vein

D. Posterior tibial vein

Answer: C. The great saphenous vein is the longest vein in the body. It starts on the medial top of the foot, passes in front of the medial malleolus, runs up the medial leg and thigh in the fat under the skin, and drains into the femoral vein.

Why the others are wrong

- **A** The femoral vein is deep, beside the femoral artery.
- **B** The popliteal vein is deep, behind the knee.
- **D** The posterior tibial vein is deep, running with the posterior tibial artery behind the medial malleolus, not in front of it.

QUESTION 9 · COMPETENCY 127

A tight cast around the neck of the fibula weakens the muscles that lift the foot. Which nerve is injured?

A. Tibial nerve

B. Common fibular nerve

C. Femoral nerve

D. Obturator nerve

Answer: B. The common fibular nerve wraps around the neck of the fibula just under the skin, and its deep branch supplies the anterior leg muscles that dorsiflex the foot. Pressure there causes foot drop.

Why the others are wrong

- **A** The tibial nerve runs deep in the back of the leg and supplies the posterior leg muscles, which point the foot down.
- **C** The femoral nerve supplies the anterior thigh (quadriceps and sartorius) and never reaches the fibula.
- **D** The obturator nerve supplies the adductors of the medial thigh.